

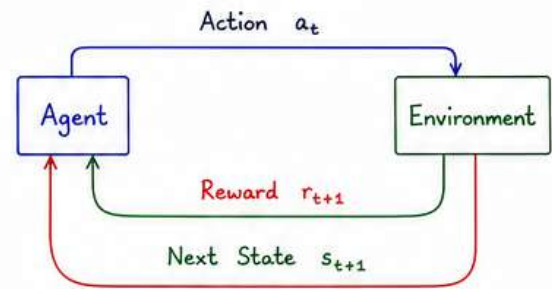
ELEMENTS OF REINFORCEMENT LEARNING (RL)

In RL, an **agent** interacts with an **environment** over discrete time steps to learn a policy that maximizes cumulative **reward**.

1. Elements of RL

- ① **Agent** : The learner or decision maker.
- ② **Environment** : Everything the agent interacts with.
- ③ **State (s_t)** : The current situation of the environment.
- ④ **Action (a_t)** : The choice made by the agent from available actions.
- ⑤ **Reward (r_{t+1})** : Immediate feedback received after taking an action.
- ⑥ **Next State (s_{t+1})** : The new state reached after action a_t .
- ⑦ **Policy (π)** : The strategy that the agent follows to choose actions. $\pi(a|s) = P(a_t = a | s_t = s)$
- ⑧ **Value Function** : A function that estimates the expected cumulative reward from a state or state-action. $V(s), Q(s, a)$
- ⑨ **Episode** : A sequence of interactions that starts from an initial state and ends in a terminal state.
- ⑩ **Discount Factor (γ)** : Determines the importance of future rewards. $0 \leq \gamma < 1$

RL Interaction Loop



Typical Notation

- s_t : state at time t
- a_t : action at time t
- r_{t+1} : reward received after taking a_t
- s_{t+1} : next state
- π : policy
- γ : discount factor ($0 \leq \gamma < 1$)

2. Elements of RL and Tic-Tac-Toe Example

Element	Explanation in RL	Tic-Tac-Toe Example																		
① Agent	The decision maker that learns by interacting with the environment.	The player (X) who wants to win the game.																		
② Environment	Everything outside the agent that responds to the agent's actions.	The Tic-Tac-Toe board and the opponent (O).																		
③ State (s_t)	The current situation of the board.	Any valid board configuration. <table border="1" style="display: inline-table; vertical-align: middle;"> <tr><td>X</td><td>O</td><td>X</td></tr> <tr><td>-</td><td>X</td><td>-</td></tr> <tr><td>O</td><td>-</td><td>-</td></tr> </table>	X	O	X	-	X	-	O	-	-									
X	O	X																		
-	X	-																		
O	-	-																		
④ Action (a_t)	The move (cell) chosen by the agent.	Placing 'X' in any empty cell (e.g., top-left corner). <table border="1" style="display: inline-table; vertical-align: middle;"> <tr><td>1</td><td>2</td><td>3</td></tr> <tr><td>4</td><td>5</td><td>6</td></tr> <tr><td>7</td><td>8</td><td>9</td></tr> </table>	1	2	3	4	5	6	7	8	9									
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⑤ Reward (r_{t+1})	Immediate feedback after the move. +1 for win, -1 for loss, 0 otherwise.	+1 if X wins, -1 if O wins, 0 for draw or non-terminal states.																		
⑥ Next State (s_{t+1})	New board after the action is applied.	Board after X places in a cell. Before Action <table border="1"> <tr><td>X</td><td>O</td><td>X</td></tr> <tr><td>-</td><td>X</td><td>-</td></tr> <tr><td>O</td><td>-</td><td>-</td></tr> </table> → After Action <table border="1"> <tr><td>X</td><td>O</td><td>X</td></tr> <tr><td>X</td><td>X</td><td>-</td></tr> <tr><td>O</td><td>-</td><td>-</td></tr> </table>	X	O	X	-	X	-	O	-	-	X	O	X	X	X	-	O	-	-
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⑦ Policy (π)	Strategy to choose the best move in each state.	A rule like: "Always try to win, otherwise block opponent, otherwise take center."																		
⑧ Value Function	Estimates how good a state or state-action is.	$Q(s, a)$ gives value of making a move a in board state s .																		
⑨ Episode	A complete game from start to end.	From an empty board until win, loss or draw. <table border="1" style="margin-right: 10px;"> <tr><td>-</td><td>-</td><td>-</td></tr> <tr><td>-</td><td>-</td><td>-</td></tr> <tr><td>-</td><td>-</td><td>-</td></tr> </table> → ... → <table border="1"> <tr><td>X</td><td>O</td><td>X</td></tr> <tr><td>X</td><td>X</td><td>O</td></tr> <tr><td>O</td><td>X</td><td>O</td></tr> </table>	-	-	-	-	-	-	-	-	-	X	O	X	X	X	O	O	X	O
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⑩ Discount Factor (γ)	Balances immediate and future rewards. Higher γ → cares more about future.	γ close to 1 means the agent tries to win in the long run, not just immediate moves.																		

Tic-Tac-Toe RL Setup

- **States (S)**: All possible board configurations. ($3^9 = 19683$ possible states including illegal ones)
- **Actions (A)**: All empty cells (1 to 9).
- **Rewards (R)**:
 - +1 → X wins
 - 1 → O wins
 - 0 → Draw / non-terminal state
- **Terminal States**: Win/Loss/Draw states.
- **Goal**: Learn an optimal policy to always win (or at least draw).

Summary: RL involves an agent learning through trial and error by interacting with an environment. Tic-Tac-Toe is a small, finite MDP that clearly illustrates all the elements of RL.

